Electrical Power and Energy Systems 145 (2023) 108698

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Scientific mapping of optimisation applied to microgrids integrated with renewable energy systems

Kawakib Arar Tahir [a], Montserrat Zamorano [b], Javier Ordóñez García [a,*]

[a] Department of Engineering Construction and Project Management. ETS Ingeniería de Caminos, Canales y Puertos, University of Granada, Campus Fuentenueva s/n, 18071 Granada.

[b] Department of Civil Engineering, ETS Ingeniería de Caminos, Canales y Puertos, University of Granada, Campus Fuentenueva s/n, 18071 Granada, Spain

ARTICLE INFO

Keywords:
Renewable energy sources
Microgrid
Optimisation
Bibliometric analysis
Science mapping analyses
SciMAT

ABSTRACT

One of the most essential factors affecting socio-economic growth is energy. In addition to rising fuel prices, climate change has led to an interest in finding alternative sources of energy. Renewable energy sources, especially those based on hybrid systems, are very popular as alternative sources of energy. The integration of hybrid energy systems into the main grid has permitted the use of microgrids that operate in both on- and off-grid modes. The literature has paid a lot of attention to the optimisation of hybrid microgrid systems. To get useful and fair information for future research trends, this study evaluates three major themes (microgrids, renewable energy, and optimisation). Through a comprehensive evaluation of the literature on hybrid microgrid optimisation and a review based on a bibliometric analysis of 2,307 Scopus records, this work employs SciMAT software to assess the state of this research area. Additionally, hidden themes and their evolution in this area were discovered from 2005 to 2021, and strategic diagrams were made to illustrate the thematic development and performance metrics throughout the course of three time periods. The findings show that this scientific subject is constantly evolving, which is supported by the significant increase in research papers, especially in the recent five years. Since 2016, studies have mostly focused on microgrids and multi-objective optimisation. In the first period, studies concentrated on smart grids and optimisation. A trend toward employing these energy systems as an alternative to conventional grids may be seen by the rise in research publications discussing microgrids that are integrated with renewable energy sources. The utilization of microgrids still faces various challenges, the most frequent of which are stakeholders, technical and financial challenges, and unpredictable renewable energy sources. This report provides a summary of the existing situation and identifies exciting new research opportunities.

1. Introduction

A reliable, affordable electricity supply is vital in today's economies. At the same time, rising oil prices and the desire to combat climate change are reshaping energy systems worldwide. Researchers have investigated alternative fuels and other solutions due to rising oil costs, particularly following the 1973 oil crisis and the 1991 Gulf War, the restricted geographic availability of oil, and the enforcement of harsher government laws on exhaust emissions [1]. To address climate change, renewable energy sources (RESs) are crucial for supplying clean, sustainable energy. In 2020, the world generated 29 % of its electricity from renewable sources, an increase of 2 % from the year before [2]. In 2020, the amount of renewable electricity generated increased by 7 %, with over 60 % coming from wind and solar photovoltaic (PV) technologies. The third-largest renewable energy technology after onshore wind and hydropower, PV contributed 3.1 % to the world's electricity production [3].

The graph in Fig. 1 demonstrates that the cost of wind and solar energy technologies decreased significantly from 2010 to 2020. During this decade, the cost of PV electricity decreased by 85 %, that of CSP by 68 %, that of onshore wind by 56 %, and that of offshore wind by 48 % [4].

RESs are an important alternative energy source, but their dependence on environmental variables like solar radiation and wind speed is a problem. As a result, individual energy sources can't provide a consistent power supply to meet demand because of their unpredictable

* Corresponding author.
E-mail address: javiord@ugr.es (J. Ordóñez García).

https://doi.org/10.1016/j.ijepes.2022.108698
Received 6 June 2022; Received in revised form 4 September 2022; Accepted 28 September 2022
Available online 8 October 2022

and irregular nature. However, RESs can be combined to create hybrid systems that are more dependable and environmentally friendly[5]. Hybrid RESs (HRESs) can operate as isolated systems or can be connected to the grid and are made up of either a single source of renewable energy and/ or a single source of conventional energy, or more than one renewable source with or without conventional energy sources [6]. These types of RES are referred to as distributed energy resources (DERs), and the generation is known as distributed generation (DG) [7]. The fundamental components of HRESs are loads, AC/DC converters, and energy sources (including renewable and/ or non-renewable energy sources, as well as energy storage systems), as depicted in Fig. 2. Studies show that PV and wind energy combinations are becoming more prevalent because of their complementing natures [8,9]. The fluctuation of these supplies in reaction to external circumstances is one of the major obstacles that could be overcome by the deployment of energy storage devices. The optimal size of house installations consisting of PV sources and lithium-ion batteries, and the influence of these installations on the main grid, were reviewed in [10] by de Oliveira e Silva and Hendrick. Their findings suggest that 30 % of the electrical supply required for self-sufficiency may be supplied by PV alone. More than 40 % self-sufficiency necessitates energy storage, which drives up the price of these facilities. To overcome the discrepancy between the electricity produced by PV systems and the demand from home appliances, Schram et al. in [11], focused on the techno-economic aspects of PV-coupled batteries in order to address the mismatch between the electricity generated by PV systems and the demand from household appliances. Sharma et al. [12] addressed the challenge of minimising the MG operation cost, and a cost-based formulation was used to identify the best battery size. A grey wolf optimisation (GWO) scheme was used to tackle the issue under various constraints. Several studies have proposed optimisation techniques for resolving issues in renewable energy (RE) systems. Recent years have seen an upsurge in research utilizing optimisation approaches to address RES difficulties, notably with regard to solar and wind systems [13,14]. In an HRES, a diesel power generator may be used as backup due to the unpredictability of RE supplies. The optimisation of a hybrid PV-wind-battery-diesel system, which is typically employed as a stand-alone HRES in subsequent studies, is another thoroughly researched arrangement.

In order to establish the optimal design of an HRES system in an unpredictable environment, Monte Carlo simulation and simulation optimisation techniques were examined by Chang and Lin [15]. In [16], Guangqian et al. demonstrated the effectiveness of two metaheuristic algorithms, harmony search and simulated annealing, to obtain the optimum design of a standalone HRES. In [17], the effects of safety parameters on the reliability and cost were explored in different HRES scenarios to find the optimum design using Monte Carlo simulation. In [18], Ogunjuyigbe et al. used a genetic algorithm (GA) to implement a tri-objective design for a stand-alone HRES. The goal of this study was to lower the life cycle cost (LCC), carbon dioxide emissions, and dump energy for a typical residential complex. The authors of [19] introduced an optimisation module based on a multi-objective GA, an uncertainty module that created uncertainty scenarios using the Latin hypercube sampling method and Monte Carlo simulation, and a simulation module, which simulated the power system under the actual operating conditions.

The integration of HRESs into the utility grid has paved the way for the use of microgrids (MGs). An MG offers a self-contained system made up of distributed energy resources that can operate in an islanded condition after a grid outage [20]. MGs are “electrical distribution systems that contain loads and distributed energy resources (such as distributed generators, storage devices, or controllable loads) that may function in a controlled, coordinated manner whether connected to the main grid or islanded mode” [21].

In order to maximise system reliability and minimise system costs and emissions, the design of a hybrid MG system (HMGS) takes into account a number of variables, such as component size, the mode of operation, the choice of location, and the selection of material. Investors may find RE technology more alluring because of the potential for providing reliable, affordable electricity [22,23,24].

The design, optimisation, operation, and control of HMGSs have been the subject of extensive research in recent years [25]. One of the aforementioned objectives, the optimisation of an HMGS, has received a lot of attention in the literature. Fig. 3 shows a diagram of the procedures needed to describe and design an optimisation problem.

Although single-objective optimisation approaches have been used in the majority of studies, there has been a growing trend in recent publications to apply a multi-objective optimisation strategy. For the study’s sites, the authors of [22,23] suggested a multi-objective particle swarm optimisation (MPSO) scheme to select the optimum configuration of a PV/wind/diesel/battery HMGS and the optimal component sizes for the locations used in the study. In [26], Kharrich et al. employed three multi-objective optimisation methods, namely MOPSO, pareto

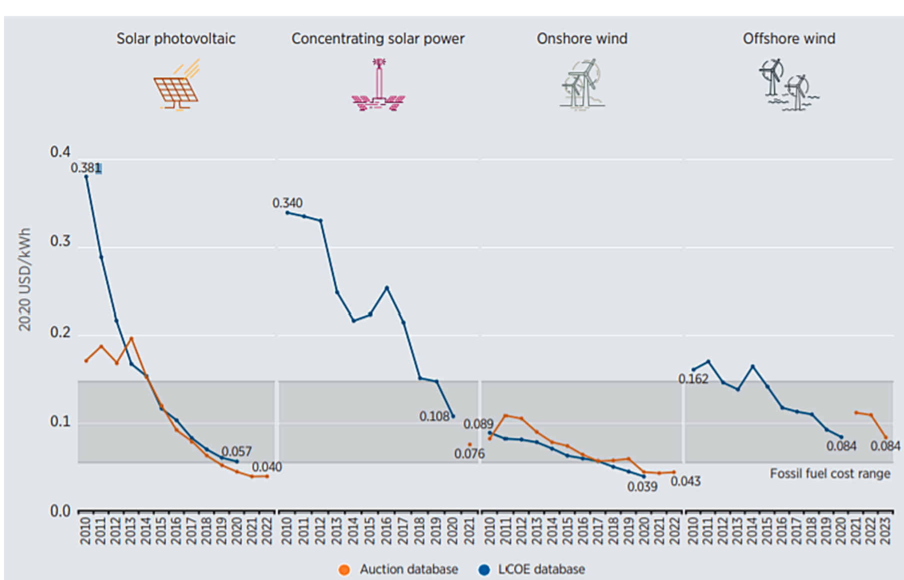

Fig. 1. Global weighted average and power purchase agreements for solar PV, onshore wind, offshore wind, and CSP, 2010–2023 (source: IRENA Renewable Cost Database).

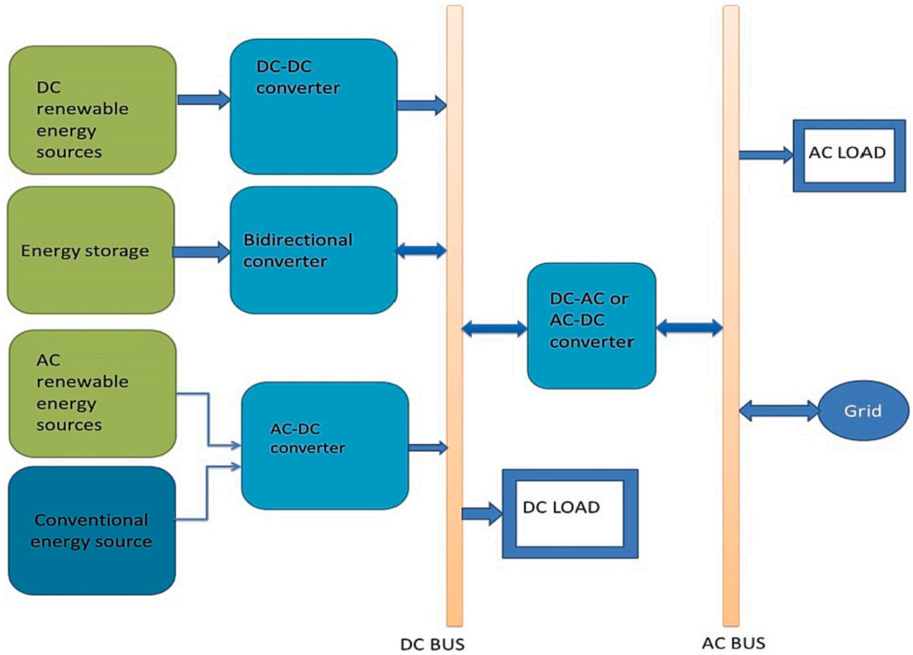

Fig. 2. Essential elements of a hybrid energy system.

Fig. 3. Schematic diagram of the required phases to build optimisation problem.

envelope-based selection algorithm II (PESA II), and strength pareto evolutionary algorithm 2 (SPEA2), to design a PV/wind/diesel/battery HMGS. Dougier et al. [27] suggested an MG design to establish compromises between technological, economic, and environmental goals using a non-weighted multi-objective optimisation process based on a GA called non dominated sorting genetic algorithm II (NSGA-II).

The method of analysis and the problem formulation have had an impact on the strategies used for the size optimisation of HMGSs. Due to the wide range of disciplines and approaches involved, it is therefore hard to provide a single entry point to this issue. Furthermore, obtaining useful and unbiased information for future research is challenging due to the narrow focus of this field of study and the quick evolution of its problems. Recent bibliometric analysis research on MGs have focused on topics like optimal battery energy storage systems [28], control strategies [29], and optimal energy storage system [30] algorithms [30]. Despite the fact that there are review articles on HMGs optimisation [13,31–33], there are no bibliometric studies on this research topic. There is, thus, a need for comprehensive review to make the integration of these contributions easy and to provide critical perspectives.

A bibliometric study offers a macroscopic picture of a sizable body of scientific literature [34] as well as objective standards for judging the work of researchers [35]. In 1969, Alan Pritchard developed the idea of bibliometric analysis. But bibliographic studies have been used since the nineteenth century in some fields of study [36,37].

The two main approaches used in bibliometric analysis are performance analysis and scientific mapping. These techniques offer a visual representation of the relationships between the important disciplines, fields, particular research articles, and authors [38], and they can be used to study bibliographic material from both an objective and a quantitative perspective [39]. In this paper, the authors use a dual integrated analysis, a systematic literature review (SLR), and a science mapping analysis to examine the present body of knowledge on optimisation research as applied to HMGSs. The current research will add to existed knowledge by analysing and highlighting trends and pattern in optimisation research, distinguishing research subjects, identifying potential research fields by mapping researcher networks. To accomplish this, the following specific goals were established: (i) a quantitative analysis based on a systematic review of the literature (SLR); (ii) a bibliometric analysis based on performance analysis and science mapping; and (iii) a comparative analysis of case studies of optimisation applied to MGs integrated with RESs.

2. Materials and method

Using a dual integrated analysis, the study's objectives were achieved (Fig. 4). The following actions were taken during this analysis: (i) a systematic literature review (SLR) of bibliographic records of optimisation as applied to MGs integrated with RESs; and (ii) a bibliometrics analysis based on performance analysis and science mapping. The following sections go into further depth about each of these phases.

2.1. SLR of bibliographic records on optimisation applied to MGs integrated with RESs

In this part, the authors describe how they searched the pertinent corpus of literature using simple, understandable search options and predetermined selection criteria. A systematic approach known as an SLR adheres to pre-established qualifying criteria while incorporating all relevant evidence [40]. This method reveals the gaps, lessens study bias, and generates reliable data from which judgments can be made

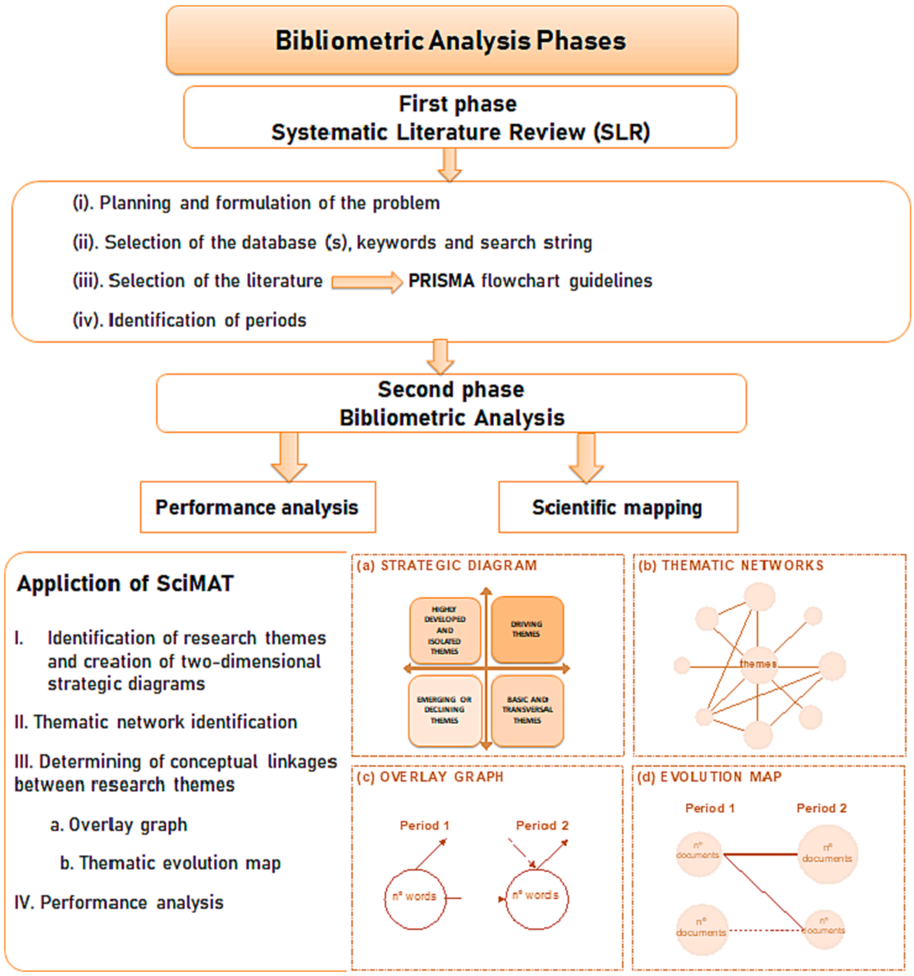

Fig. 4. Flowchart of bibliometric analysis phases.

[41]. An SLR adheres to a protocol that specifies the search string, approach, exclusion standards, and data extraction techniques [42]. As a result, some authors have utilized SLRs in their studies and the numerous procedures necessary to create a reproducible, scientific, and transparent research process [40,43–47]. This paper's SLR is based on the suggestions in [48] (Fig. 4), which were gathered utilizing the stages below:

(i) Planning and formulation of the problem: In this section, the research questions are chosen in accordance with the objectives, the criteria and procedures used to select the acceptable bibliographic records, the system for rejecting results that are not relevant, and the anticipated outcomes.

(ii) Selection of the database(s), keywords and search string: This stage identifies the bibliographic database(s), keywords, and search term. Determining the keywords and using the right search term are crucial. The list of keywords should be extensive enough to include a variety of studies, yet sufficiently constrained to only identify publications that are pertinent to the subject.

(iii) Selection of the literature: In this stage, The PRISMA (reporting systematic reviews and *meta*-analysis) flowchart's instructions are followed for choosing the publications that are pertinent to the study's subject [49]. The information required to accomplish the study's objectives is found in papers that are relevant.

(iv) Identification of periods: This decision is relied on several elements, the most crucial which are the elements and variables of the study, the number of research studies about such topic, and the changes happening in this scope.

2.2. *Bibliometric analysis: Scientific mapping and performance analysis*

In the second stage, a bibliometric analysis is achieved after applying the SLR in the first stage. It has two combined analysis: it includes a performance analysis and a scientific mapping. The performance analysis evaluates the impact of citations on scientific outputs, whereas the scientific mapping extracts the conceptual and intellectual structure of the research in this area, its evolution, and dynamic aspects. The aim of a bibliometric analysis is to represent the relationships between various specialisations, disciplines, documents, individual authors, and documents.

A bibliometric analysis is carried out in this work using the Scientific Mapping Analysis Software Tool (SciMAT v1.1.049). SciMAT relies on a mutual word analysis [50] and the h-index [51]. The h-index, which is based mostly on the number of studies and citations of a given researcher's output, can be used to assess the quality of a publication. Hirsch [52] defined the h-index as follows: "A scientist has index h if h of his or her number of published (Np) papers have at least h citations each and the other (Np - h) papers have $\leq$ h citations each". From preprocessing to presenting the results of the scientific mapping, SciMAT offers methodologies, algorithms, and measurements for each step of a typical scientific mapping workflow [53,54]. Construction, urban planning, waste management, energy, and information technology are only a few of the fields in which this program has been successfully used [55–66]. The workflow of this tool is based on the following phases.

I. Identification of research themes and creation of two-dimensional strategic diagrams: The software initially creates an equivalence index [67], before using the single-centre approach [68] to identify the topics that are the most pertinent. A strategic strategy is then created for each phase. The centrality and density concepts [67] provide the foundation for the diagrams. The relationship between the main study issue and related research topics is crucial. According to density, which is used to gauge the topic's level of development, the internal coherence of all relationships among the terms that define a topic is measured [54,52]. In order to illustrate the four different categories of study topics, the diagrams are divided into four quadrants (Fig. 4):

- **Driving themes**: These are well-developed and crucial issues in the scientific discipline, located in the upper right quadrant. High density and strong centrality are characteristics of the themes that are essential to the growth of the study field.
- **Highly developed and isolated themes**: These are topics that are distinct from the other themes but are internally very well developed. They are displayed in the upper left quadrant and are specialized themes in ancillary areas of the research field.
- **Emerging or declining themes:** These are concepts that lack development and relevance and are displayed in the lower-left quadrant. They could gain importance or lose it. This will be decided in the following time frame.
- **Basic and transversal themes:** These themes are major to the research field, but aren't fully formed yet. They can be seen in the lower right-hand corner of the quadrant.

II. Thematic network identification: These are used to supplement the strategic diagrams and examine the links and linkages between the topics and keywords in the strategic diagrams. According to Fig. 4, each thematic network is given a name that corresponds to the theme's most crucial term. Here, several keywords are linked; the size of the circle is related to the number of documents that correspond to each keyword, and the thickness of the link between two circles is proportional to the equivalence index.

III. Determining conceptual linkages between research themes: The inclusion index [60] pinpoints conceptual relationships between research themes throughout various eras. The firmness of association between themes is also determined. The following two kinds of graphs are utilized (Fig. 3):

a. In the **overlay graph**, the quantity of words shared between the two periods is indicated by a horizontal arrow. The upper arriving arrow indicates the number of new words in the second period, while the upper outgoing arrow indicates the number of words that vanish in this time.
b. **A thematic evolution map** shows that the linked theme and the primary item are connected by solid lines. A dotted line denotes that the themes share non-main item elements. The volume of each sphere is proportional to the number of published documents, and the thickness of each edge is proportional to the inclusion index.

IV. Performance analysis: Both quantitative and qualitative standards are used to the proportional contribution of study participants to the overall research field. It's also utilized to state the most productive, notable and high effect subfields utilizing bibliometric indicators, such as, the number of published citations, articles and various kinds of h-index.

3. Results and discussion

The outcomes of the steps are described in the previous section and are analysed in Figs. 4–8 and Tables 1–4.

3.1. SLR of bibliographic records on optimisation applied to MGs integrated with renewable energy systems

This work aims at examining the present state of knowledge on the optimisation of HMGSs. To perform this goal, the following research questions (RQ) were established:

RQ1: What is the present state of research in this scope?

RQ2: What are the key concepts that define the research field?

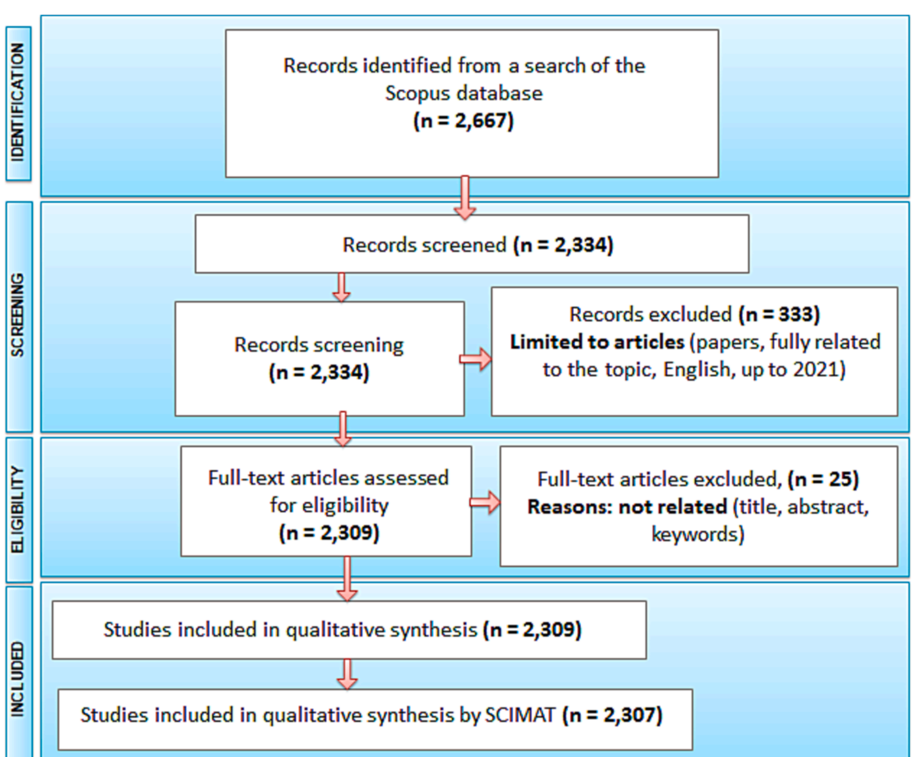

Fig. 5. PRISMA flowchart.

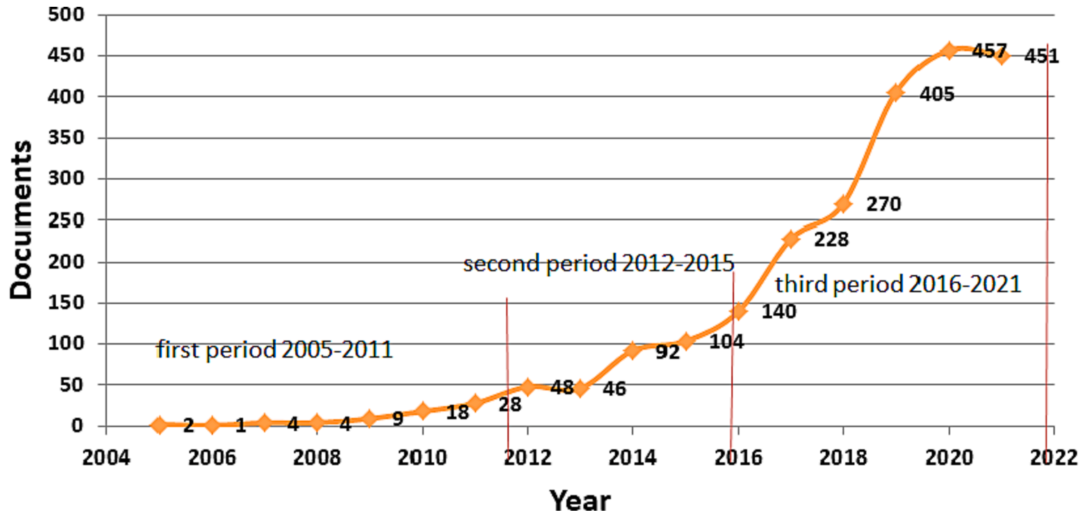

Fig. 6. Number of documents per year.

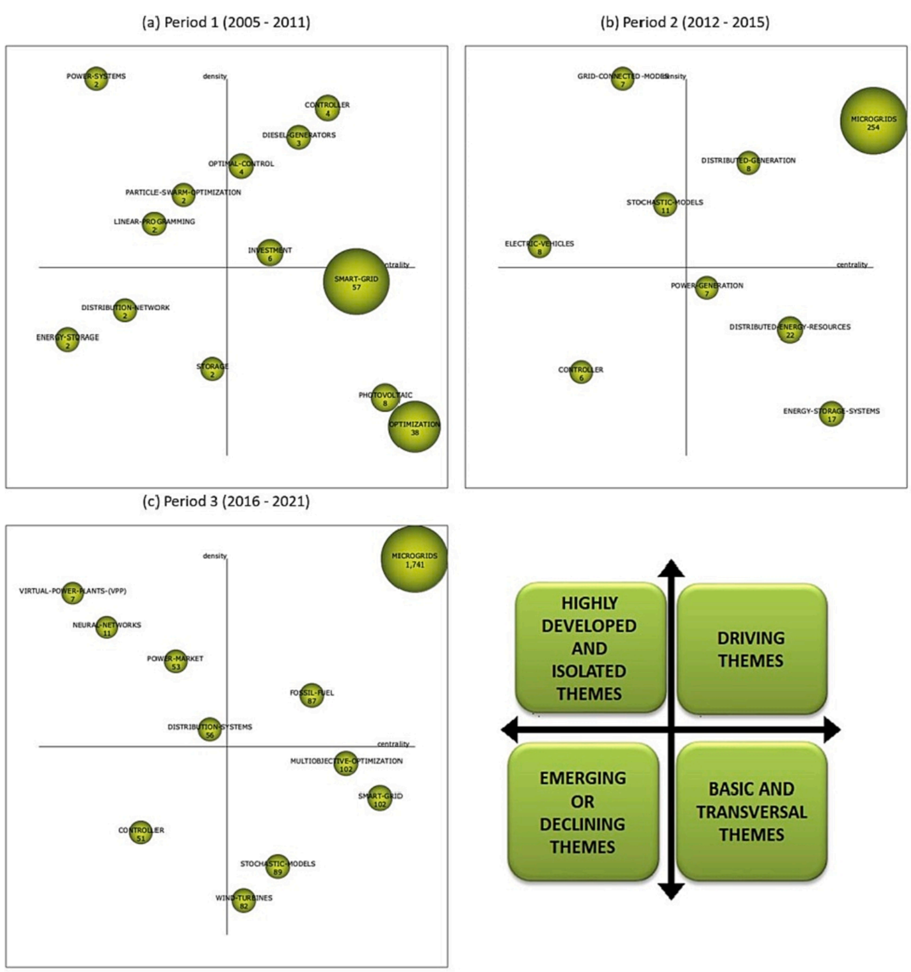

Fig. 7. Strategic diagrams for (a) period 1; (b) period 2; (c) period 3.

RQ3: What are the present issues in this field of research?

RQ4: What are the most turning points and pressing concerns in terms of the research subject?

RQ5: What are the most pressing concerns in the field?

RQ6: What are the limitations of existing research?

RQ7: What are the most influential works in this area of research?

RQ8: Who are the most productive authors in the research area?

This inquiry made use of the SCOPUS database. This database contains a large number of high-impact scientific and technological publications written by scholars from all over the world who specialize in various fields. MGs, renewable energy, and optimisation were the three main areas on which the review was centered. The following keywords

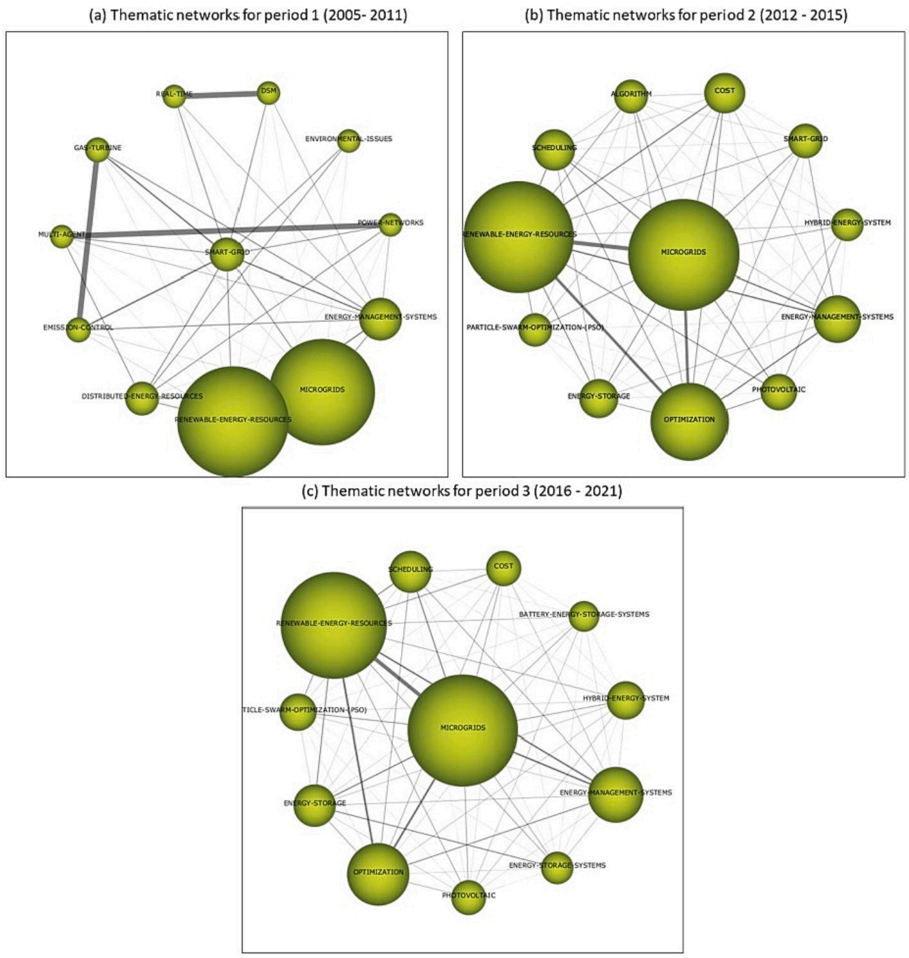

Fig. 8. Thematic networks for (a) period 1; (b) period 2; (c) period 3.

were then applied to each topic using an advanced search of SCOPUS. Finally, an advanced search of SCOPUS was conducted, and the following keywords were assigned to each topic.

The structure of the search was as follows: (TITLE-ABS-KEY (“microgrid” AND “renewable energy” OR “hybrid energy”) AND TITLE-ABS-KEY (“optimization” OR “optimisation” OR “techno economic evaluation”)) AND (EXCLUDE (DOCTYPE, “bk”) OR EXCLUDE (DOCTYPE, “ed”) OR EXCLUDE (DOCTYPE, “tb”) OR EXCLUDE (DOCTYPE, “Undefined”) OR EXCLUDE (DOCTYPE, “ch”) AND (EXCLUDE (SUBJAREA, “MEDI”) OR EXCLUDE (SUBJAREA, “BIOC”) OR EXCLUDE (SUBJAREA, “NEUR”) OR EXCLUDE (SUBJAREA, “ARTS”) OR EXCLUDE (SUBJAREA, “IMMU”) OR EXCLUDE (SUBJAREA, “PHAR”)) AND (EXCLUDE (PUBYEAR, 2022)) AND (LIMIT-TO (LANGUAGE, “English”)) AND (EXCLUDE (LANGUAGE, “Chinese”) OR EXCLUDE (LANGUAGE, “Japanese”)).

From the Scopus database, 2,667 bibliographic records were retrieved using a procedure based on the PRISMA flowchart. 333 records were removed altogether after the exclusion criteria were applied (only article papers, entire documents connected to the research topic, and materials in English were considered). The title, abstract, and keywords were used to further screen the remaining 2,334 results and a further 25 records were omitted as they did not evoke to the themes of the review. After the final elimination process, 2,307 relevant papers remained for the study (Fig. 5).

In order to investigate trends in publication patterns, a time window of 2005–2021 was employed, which was divided into three periods depending on the number of papers selected and the pertinent milestones:

First period (2005–2011): 66 papers. The US Department of Energy (DOE) organized the first workshop around this time to determine potential areas for MG research and development. The price of PV modules dropped dramatically throughout that time period. In 2011, the minimal cost per watt dropped below $1.

Second period (2012–2015): 290 papers. 2015 saw a number of occasions that promoted the growth of renewable energy, including the approval of the United Nations Development Goals (SDGs) (2015) and the Paris Conference of the Parties. In terms of the SDGs, target 7.2 of goal 7 (affordable and clean energy) [69]. The major goal of the Paris Climate Conference (COP) was to improve the global response to the threat of climate change by keeping global temperature increases well below the limit of 2 °C above pre-industrial levels in this century. Generating energy from renewable sources was a critical factor in achieving this aim.

Third period (2016–2021): 1,951 In 2016, the price per watt of solar panels dropped significantly, as shown in Fig. 1. Over this time, there was also an upsurge in research. The need to combat climate change and prevent growing dependent on fossil fuels, which has become a geostrategic threat, is likely to cause RE output to increase in the next years.

Fig. 6 depicts the time distribution of the final 2,307 publications. The exponential increase in the number of related articles shows that this field is continually growing, and will continue to grow in the coming years. The number of research articles in which optimisation approaches are applied to solve the challenges of HMGS has risen considerably since 2005, indicating a definite increase in research interest in this area. The number of research articles also rose considerably after 2016,

Table 1
Performance measures for each theme.

Period 1 (2005–2011)	No. of documents	h-index	No. of citations	Centrality	Density
Controller	4	4	437	168.45	128.33
Investment	6	5	160	115.35	70.6
Diesel-generators	3	3	87	124.16	118.75
Smart-grid	57	25	3,458	212.45	62.32
Optimal-control	4	4	362	104.97	113.13
Optimization	38	22	2,870	278.66	18
Photovoltaic	8	6	247	219.47	29.86
Power-systems	2	2	20	22.83	231.25
Particle-swarm-optimization	2	2	95	71.13	93.06
Distribution-network	2	2	18	32.22	52.78
Linear-programming	2	2	95	35.1	83.33
Energy-storage	2	2	26	20.1	45.83
Storage	2	2	94	75.61	37.5
Period 2 (2012–2015)	**No. of documents**	**h-index**	**No. of citations**	**Centrality**	**Density**
Microgrids	254	57	11,066	145.19	51.48
Distributed-energy-resources	22	14	1,487	35.71	7.67
Energy-storage-systems	17	9	471	40.35	5.45
Stochastic-models	11	4	281	12.67	14.38
Power-generation	7	6	684	24.57	8.07
Distributed-generation	8	7	547	27.85	30.54
Grid-connected-modes	7	4	176	12.47	84
Electric-vehicles	8	6	504	10.2	10.29
Controller	6	4	489	10.82	6.07
Period 3 (2016–2021)	**No. of documents**	**h-index**	**No. of citations**	**Centrality**	**Density**
Microgrids	1,741	73	24,354	131.4	45.83
Controller	51	12	517	9.64	2.81
Smart-grid	102	32	3,094	29.61	2.88
Fossil-fuel	87	23	1,739	26.79	4.15
Power-market	53	14	987	17.69	4.49
Stochastic-models	89	24	1,707	25.27	2.34
Multiobjective-optimization	102	21	1,773	28.06	3.05
Wind-turbines	82	23	1,779	21.15	1.93
Distribution-systems	56	18	1,156	18.34	3.69
Virtual-power-plants-(vpp)	7	4	93	5.28	22.59
Neural-networks	11	6	117	7.88	7.52

* Number of citations up to 14/1/2022.

particularly for wind and solar energy systems, coinciding with the adoption of the SDGs, the COP, and DOE workshops. Some of these optimisation methods were based on traditional techniques, and some researchers have successfully tackled multi-objective problems involving RESs.

3.2. Bibliometric analysis: Science mapping and performance analysis

The following graphs are analysed and discussed in this section: strategic diagrams (one for each period); critical thematic networks; an overlay graph; and a thematic evolution map. In addition, the evolution of the time horizon for the documents, the number of citations, the most cited authors, and the publications themselves are numerically and qualitatively examined.

Table 2
A comparative study of the findings of the strategic diagrams for the three periods.

	Comparative based on	First period (2005–2011)	Second period (2012–2015)	Third period (2016–2021)
1.	Research themes and the number of publications	From 66 papers, 13 research themes recognized	In 290 papers, 9 research themes were observed	1,951 papers contained information on 11 research themes. Most articles were published over this time period, demonstrating the significance of the subject and its ongoing progress
2.	The highlighted themes	The themes of smart grid and optimisation discussed in a large number of documents	Given the volume of citations and the number of articles addressing MGs and Distributed-energy-resources themes, substantial attention was paid to these areas	The driving themes have been recognized as MG and fossil fuels. The findings reveal a strong bias in the volume of citations and articles addressing MG
3.	The emergence of themes (MG, RESs, and Optimisation)	MGs theme does not emerge in this period while the other major themes and its hidden related topic like optimisation (Linear-programming, PSO) and hybrid energy (DG, PV, storage, energy-storage) are emphasized	MGs and Distributed-energy-resources have focused as driving themes. It should be noted here that this period was the beginning of the focus on utilizing MGs	The themes (MGs and Multi-objective-optimisation) have received high attention since they were one of the results and findings of the DOE workshop in 2011, which suggested the development of multi-objective optimisation and utilizing it as a tool for economic and design analysis of Microgrid [21]
4.	High cited documents	The documents with more than 100 citation [70–75] focused on optimisation, optimal control, and controller	The articles [76–83] with the most citations examined HMGs' technological, economic, and environmental aspects	In this period, the various aspects of evaluating HMGs, including feasibility, environmental impact, sizing, and technicalities, as well as the usage and development of various optimisation approaches, were discussed

3.2.1. Strategic diagrams

Three strategic diagrams were made for the three time periods under consideration (2005–2011, 2012–2015, and 2016–2021) (see Fig. 7), where the size of the circle represents the number of papers linked to each research topic that have been published. The performance measurements for each topic and time period are shown in Table 1 in terms of the quantity of documents, the h-index, the centrality and density values, and the number of documents. Below is a breakdown of the results for each time period.

Table 3
Prominent journals contributing to the research field.

Name	No. of articles from the journal	No. of citation indexes of the journal	Title of the most cited document	Number of citations of the document	Ref.
Applied Energy	87	4791	Microgrids energy management systems: A critical review on methods, solutions, and prospects	349	[20]
IEEE Transactions on Smart Grid	58	4489	Intelligent frequency control in an AC microgrid: Online PSO-based fuzzy tuning approach	355	[96]
Energy	63	2451	Probabilistic energy and operation management of a microgrid containing wind/photovoltaic/fuel cell generation and energy storage devices based on point estimate method and self-adaptive gravitational search algorithm	193	[97]
IEEE Transactions on Sustainable Energy	30	1823	Robust energy management for microgrids with high-penetration renewables	510	[78]
International Journal of Electrical Power and Energy Systems	60	1477	Solve environmental economic dispatch of Smart MicroGrid containing distributed generation system - Using chaotic quantum genetic algorithm	126	[82]
Energy Conversion and Management	26	1406	Efficient energy consumption and operation management in a smart building with microgrid	240	[98]
Energies	127	1336	Stochastic modeling and optimization in a microgrid: A survey	130	[99]
Renewable Energy	26	1102	Optimal sizing of PV/wind/diesel hybrid microgrid system using multi-objective self-adaptive differential evolution algorithm	209	[100]
IEEE Access	80	898	A microgrid energy management system and risk management under an electricity market environment	88	[101]
IET Generation, Transmission and Distribution	24	594	Grey wolf optimisation for optimal sizing of battery energy storage device to minimise operation cost of microgrid	139	[12]

* Number of citations up to 14/1/2022.

Table 4
Main authors contributing to the research field.

Name	No. of documents	Total citations	h-index	Title of the most cited document	Number of citations of the document	Ref.
Guerrero, J. M.	35	1350	105	Microgrids: Experiences, barriers and success factors	214	[24]
Wang, J.	35	795	80	Decentralized energy management system for networked microgrids in grid-connected and islanded modes	233	[103]
Wang, X.	35	500	6	Energy management of multiple microgrids based on a system of systems architecture	108	[104]
Zhang, Y.	31	820	9	Robust energy management for microgrids with high-penetration renewables	510	[78]
Liu, Y.	28	525	16	Distributed robust energy management of a multimicrogrid system in the real-time energy market	102	[105]
Xu, Y.	28	739	45	A two-layer energy management system for microgrids with hybrid energy storage considering degradation costs	147	[106]
Zhang, J.	27	400	45	Short-term photovoltaic solar power forecasting using a hybrid wavelet-PSO-SVM model based on SCADA and meteorological information	175	[107]
Wang, Y.	27	193	15	Optimal operation of microgrid with multi-energy complementary based on moth flame optimization algorithm	24	[108]
Li, Z.	26	406	6	A two-stage optimization and control for CCHP microgrid energy management	89	[109]
Wang, L.	25	640	11	Robust optimization for energy transactions in multi-microgrids under uncertainty	104	[110]

* Number of citations up to 14/1/2022.

- **First period (2005–2011):** Thirteen research themes were identified in the 66 papers selected in this period, according to the strategic diagram shown in Fig. 7a. Four were identified as driving themes (controllers, investment, diesel generators, and optimal control). Three were identified as highly developed and isolated themes (power systems, particle swarm optimisation, and linear programming). Three were identified as emerging or declining themes (storage, energy storage, and distribution networks), and three were identified as basic themes (smart grids, optimisation, and PV). A performance analysis for each theme, as shown in Table 1, added to the information supplied by the diagram, and it can be seen that the themes of smart grids and optimisation had the highest performance measurements. These themes had higher values for the h-index than the others, since they had more than 6,000 citations.
- **Second period (2012–2015):** The strategic diagram in Fig. 7b shows that nine research themes were found from the 290 papers for this period. Two were considered driving themes (MGs and distributed energy resources), three were considered highly developed and isolated themes (grid-connected modes, electric vehicles, and stochastic models), one was an emerging or declining theme (controllers), and three were basic themes (power generation, distributed energy generation, and energy storage systems). Two themes stood out based on the performance measures (Table 1), which were MGs and distributed energy resources. These research topics had higher values for the impact rate and h-index than the others.
- **Third period (2016–2021):** Eleven research themes were observed in the 1,951 publications selected for this period, according to the strategy diagram shown in Fig. 7c. Two were identified as driving themes (MGs and fossil fuels), four as highly developed and isolated themes (power markets, distribution systems, virtual power plants (VPPs), and neural networks), one as an emerging or declining theme (controllers),

and four as basic themes (wind turbines, multi-objective optimisation, stochastic models, and smart grids). Three themes were highlighted based on the performance measures (Table 1): MGs, smart grids, and optimisation. These topics had higher values for the h-index and impact rate than the others.

Three periods were set up for the three main themes that the study aims to assess (MGs, RES, and optimisation), and the results of the three strategic diagrams are in Table 1. Table 2 shows a comparative study of the findings of the strategic diagrams for the three periods.

The articles that received the most citations during the first period [70–75] dealt with the subject of control, optimal control and optimisation. Reference [73] discusses the use of power electronic converters and their contributions to achieve controllable, reliable, sizeable, and efficient systems, particularly with the use of RES. A bidirectional converter is used as an energy control centre for a DC nanogrid in a presumption sustainable house.

Given the volume of citations and the number of documents covering them, the second period saw considerable interest in the issues of MGs and distributed-energy-resources. It should be emphasized that the majority of the highly cited studies [76–83] covered the optimisation of HMGs' technical, economic, and environmental aspects. To execute the economic analysis and define optimal operation using the CPLEX solver on the GAME platform, a hybrid PV-WT-battery MG is proposed in ref. [83]. Stochastic dynamic programming technique was utilized to minimise the cost of PV- WT- DG- battery HMGs in [84]. Using an artificial bee colony (ABC), the optimal PV-WT-biomass-storage HMG size was reported in [85]. For combining demand response management and thermal comfort optimisation in MG integrated with RES and energy storage, a novel control method was developed in [86]. In order to design a hybrid RES microgrid, the optimal type, size, and location of DERs were examined in [81]. The Distributed Energy Resources Customer Adoption Model (DER-CAM) was then used to determine the best configuration and location, as well as the financial advantages of deploying these systems.

According to the information presented above, optimisation can be applied to every component of HMGs, including the power market [87], energy management for DERs integrated with MG [88], and optimal controller for virtual power plant integrated with MG [89]. The strategy diagram shows new topics that show the study field's current direction during the previous five years, such as using a neural network to solve an optimisation problem [90–92] and using multi-objective optimisation as a tool to assess HMGs [26,93]. It should be noted that the areas of optimisation applied to HMGs have been roughly divided into three categories: generation, control, and distribution.

3.2.2. Thematic networks

In this section, three thematic networks, one for each period, were developed to enable further exploration of the relevant subjects and future trends based on their saliency. Advanced, fundamental, and emerging research are all captured by these networks across time. Fig. 8 shows the relationships between the themes (circles), where each circle's size corresponds to the number of documents and each line's thickness to the strength of the connection between two nodes. Below is a breakdown of the outcomes for each time period.

In the first period, it was noted that the keyword "smart grids" was closely related to other keywords like "MGs," "renewable energy," "energy management," and "emission control." These are all interconnected in some way. According to the International Energy Agency (IEA), a smart grid is defined as follows: A smart grid is an electricity network that uses digital and other advanced technologies to monitor and regulate the transportation of electricity from all sources of generation in order to satisfy the various electricity needs of end users. In order to operate every component of the system as efficiently as possible, smart grids coordinate the needs and capabilities of all generators, grid operators, end users, and electricity market stakeholders, minimising costs and environmental impacts while maximising system reliability, resilience, and stability [94]. In order to increase the system's energy efficiency, improve power quality and stability, and give individual end-user locations the option of grid independence, MGs have been cited as a crucial component of a smart grid [21]. Energy management of the MG is essential to enable the most effective use of these resources in a dependable, secure, and coordinated way since the MG is a major component and permits the use of DERs as a source of energy. The theme of MGs emerges as the most significant over these periods based on these networks, as seen in Fig. 8b and c, which depict the thematic networks for the second and third periods. Throughout these periods, the theme of MGs is closely related to concepts like RE resources, optimisation, energy management, solar, and energy storage. These are all intimately related to one another and have recently been the focus of many research studies.

Based on the aforementioned findings, it is concluded that although the idea and the initial MG tests date back to the 1980 s, these have only recently begun to move from the experimental stage to the commercialisation stage, with pilot projects emerging worldwide [24]. The effectiveness of MG systems has been assessed using a range of performance models, indicators, tools, and optimisation techniques. The significance of the MG system has recently been recognized in a number of study papers. Most documents over the preceding ten years concentrated on the topics of smart grids, optimisation, and PV, while the top authors were interested in the RE investment aspects. Since 2012, there has been significant interest in resolving the problems with RE through the utilization of optimisation tools, and interest has also concentrated on the development of MGs. In the latest research, the concept of multi-objective optimisation has drawn the most interest from researchers. The most intriguing research fields in relation to renewable and sustainable energy MGs are those using heuristic methods, particularly genetic algorithms, particle swarm optimisation, Pareto-based multi-objective optimisation, and parallel processing.

3.2.3. Overlay graph and thematic evolution map

The number of outgoing and incoming keywords, the number and evolution of keywords for each period, the number and percentage of keywords that were maintained from one period to the next are all shown in Fig. 9a. Between the first and last periods, there were 406 % more keywords, going from 138 to 699. Of the 138 keywords that appeared in the first period, 75 % (104) were maintained in the second period, and 185 words were added, to give a total of 289. In the third period, 263 words (91 %) were maintained from the second period and 406 new words were added, giving a total of 699 words.

According to these findings, there are a lot of new and transitional keywords, and the number of keywords used in different times has grown. The fact that the research field of MG assessment is growing more thematically diverse and that certain keywords have come back stronger in future periods may be signs that this relatively new research subject is becoming normalizing.

Based on an examination of the roots of the topics and their interrelationships, Fig. 9b depicts the thematic progression of the research field. When the graph is examined in terms of the number of articles, it is clear that the first period saw the emergence of the theme of smart grids, which had the biggest number of publications. From there, the themes of MGs, distributed energy resources and DG during the second period.

The transition from conventional energy grids to smart grids is the result of growing concerns about the effectiveness, reliability, economics, and sustainability of electricity production and distribution [95]. In order to fulfil the rising demand for energy, minimise environmental pollution, and provide social and economic benefits in terms of sustainable development, RE resources are currently being used on a wide scale. The integration of these distributed energy sources into the utility grid paves the way for MGs [20]. The MG, which has both on-grid and off-grid modes of operation, is a novel approach to integrating RESs into a DG system [75]. The idea of an MG was developed to enable a self-

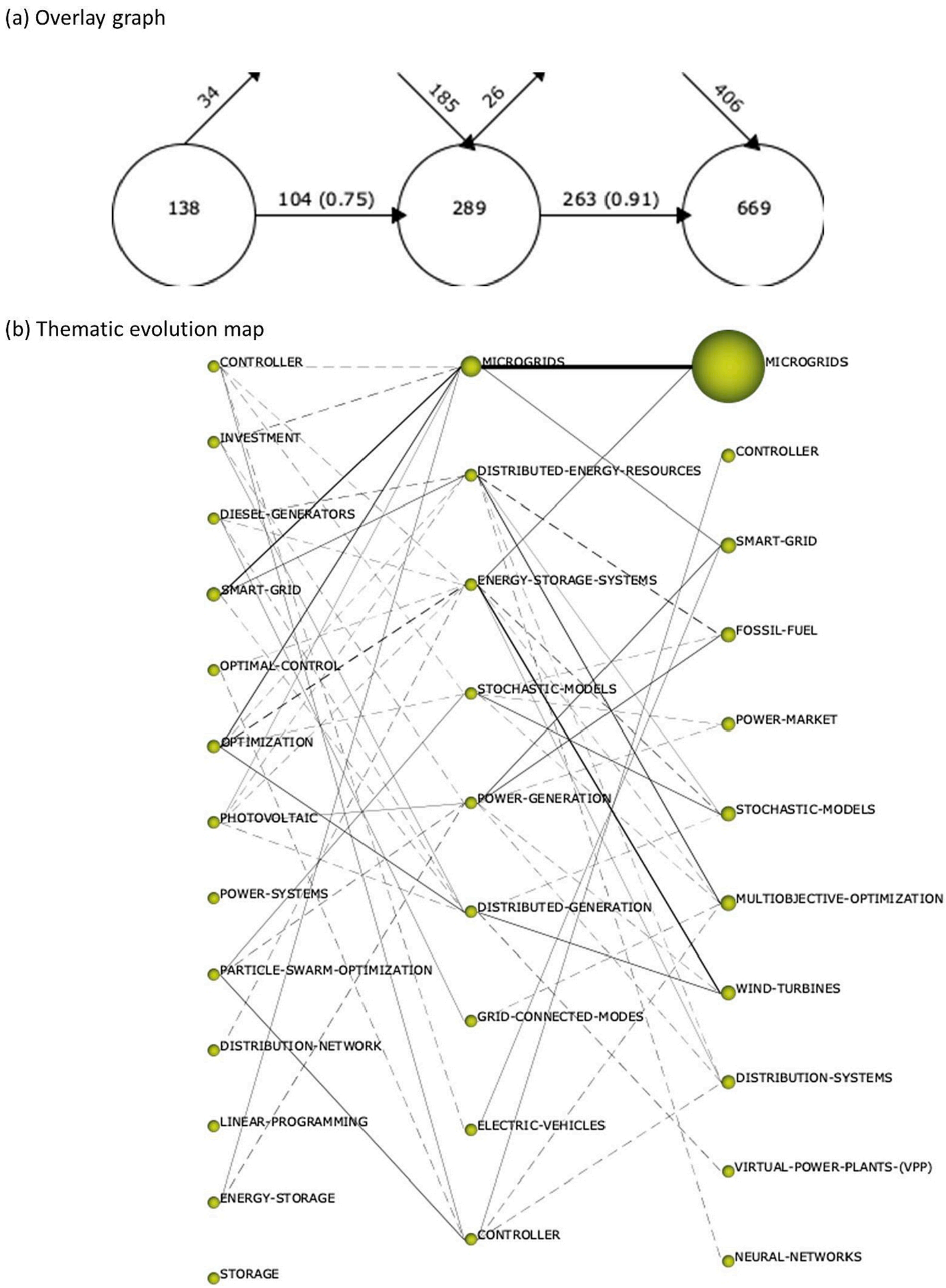

Fig. 9. (a) Overlay graph and (b) thematic evolution map.

sufficient system made up of distributed energy resources that could function in off-grid mode throughout a grid outage failure [20].

The most popular topic for papers in the second period, MGs, later became MGs and smart grids in the third period. In order to enhance power quality and reliability, raise system energy efficiency, and provide grid independence for specific end-user sites, MGs were recognized as a crucial part of smart grids. The promotion of the integration of distributed and RE resources is one of the key advantages of MGs. By identifying generation sources that are close to the demand, securing energy supplies for critical loads, maintaining local control over the quality and reliability of power, and promoting consumer involvement in the electricity supply through demand-side management, this can help to reduce peak loads and losses [21].

3.2.4. Performance analysis

951 journals were found in this study, with the majority of them concentrating on energy usage and RE, energy management and operation of MGs, improvement and optimisation of MG technologies, and technical, environmental, and economic evaluations of MGs. Table 3 displays the top ten research journals that contributed the most, with

(581) articles, or 25.18 % of the total number of documents analysed, and is sorted by the number of citations. In Table 3, the most cited article for each journal is also shown. There is no correlation between the number of research papers published and the number of citations, therefore the journals with the most published research studies are not necessarily the most significant in the area in terms of the number of citations, as shown in Table 3.

The systematic literature review of this study allowed us to identify 5,135 authors who had published articles on the topic under study. According to the total number of published documents, Table 4 lists authors with more than 10 studies that have been published. Additionally, it contains the number of citations obtained and the h-index (Hirsch index), which gauges an author's professional caliber based on how frequently their academic works are cited [102]. According to our research, Guerrero, J.M., who published 35 documents on the subject of optimisation of MG integrated with RE resources, had the highest citation count and h-index. Wang, J. and Wang, X had the same number of documents as the first author, and the majority of those documents were published solving optimisation problems based on the economic, technical, and environmental issues related with the use of MGs.

The systematic literature review ended with a study of the most frequently cited documents. 40,950 citations were made to the 2,307 documents that were examined for the review. The 10 publications with the most citations are listed in Table 5, totaling 4,243, or 10.36 % of all citations. These publications covered a variety of topics, including the development and assessment of MGs, the utilization of RE resources, and the presenting of optimisation techniques for MGs' operational and financial performance.

Table 5
Most frequently cited documents.

Title	Author(s)	Year	No. of citations	Ref.
Centralized control for optimizing microgrids operation	Hatziargyriou, N.D., Tsikalakis, A.G.	2008	797	[70]
Robust energy management for microgrids with high-penetration renewables	Zhang, Y., Giannakis, G.B., Gatsis, N.	2013	510	[78]
Future electronic power distribution systems—A contemplative view	Boroyevich, D., Cvetković, I., Dong, D., Burgos, R., Wang, F., Lee, F.	2010	481	[73]
Multiobjective intelligent energy management for a microgrid	Chaouachi, A., Kamel, R.M., Andoulsi, R., Nagasaka, K.	2013	471	[79]
Optimization in microgrids with hybrid energy systems—A review	Fathima, A.H., Palanisamy, K.	2015	380	[13]
Intelligent frequency control in an AC microgrid: Online PSO-based fuzzy tuning approach	Mitani, Y., Bevrani, H., Habibi, F., Babahajyani, P., Watanabe, M.	2012	355	[96]
Distributed intelligent energy management system for a single-phase high-frequency AC microgrid	Chakraborty, S., Weiss, M.D., Simões, M.G.	2007	351	[74]
Microgrids energy management systems: A critical review on methods, solutions, and prospects	Zia, M.F., Elbouchikhi, E., Benbouzid, M.	2018	349	[20]
Optimal bidding strategy for microgrids considering renewable energy and building thermal dynamics	Nguyen, D.T., Le, L.B.	2014	276	[111]
Control methods of inverter-interfaced distributed generators in a microgrid system	Liu, W., Chung, I.-Y., Cartes, D.A., Collins Jr., E.G., Moon, S.-I	2010	273	[75]

* Number of citations up to 14/1/202.

3.3. *Comparative analysis of case studies of optimisation applied to MGs integrated with RESs*

An analysis was carried out for the relevant SLR documents and a comparison of case studies on optimisation applied to HMGS. Hybrid energy sources are becoming more and more common in light of the current energy situation and environmental issues. This study addressed the significant advancements in research surrounding the usage of MGs integrated with RESs and the application of optimisation approaches to address issues related to the utilization of these resources. Artificial intelligence (AI) can be utilized to optimise the technical and financial components of these systems, as evidenced by the current trend in HMGSs optimisation. MGs vary greatly depending in terms of their location, components, and the aims of optimisation, which leads to a range of challenges and barriers. Four categories of the most frequent hurdles have been determined: technological, regulatory, financial, and stakeholder constraints [24]. Research on using optimisation approaches to identify the solutions for HMGSs is listed in Table 6; the studies were chosen based on the number of citations.

The goal of optimisation, as described by Nguyen and Le in [111], was to strike a balance between maximising the expected benefits of the MG in the liberalized electricity market and minimising operating costs while taking into account the users' needs for thermal comfort and other system constraints. A compact standalone PV-wind-biomass-battery hybrid system was described by Singh et al. [85], who also created a mathematical model to determine the optimal sizes of this hybrid system's components using an artificial bee colony algorithm (ABC). Their outcomes were contrasted with those of two common software applications, hybrid optimisation model for electric renewable (HOMER) and particle swarm optimization (PSO). The suggested algorithm produced superior outcomes.

Atia and Yamada looked into how load flexibility affected the component sizing of a hybrid system for a residential MG in [112]. In order to simulate varied technological constraints and uncertainty levels, they also took into account alternative operation situations. The stochastic models were created using Matlab software, and mixed integer linear programming (MILP) was employed to optimise the system design. The models were solved utilizing the CPLEX optimisation engine as deterministic problems, which required about 90 min per instance. The crucial topic of frequency management of AC MGs in the presence of disturbances, uncertainties, and load fluctuations was examined by Bevrani et al. in [96]. Adaptive control with two levels, based on a PI controller and a fuzzy system method, was used to control the frequency of the AC MG system. The PSO algorithm was employed to enhance the membership function's parameters because of the membership function's substantial reliance on the fuzzy system. Ramli et al. [100] studied the optimal sizing of a PV/wind/diesel HMGS with battery storage using a multi-objective self-adaptive differential evolution (MOSaDE) algorithm. A multi-objective optimisation approach was then used to analyse the loss of power supply probability (LPSP), the cost of electricity (COE), and the renewable factor (RF) in relation to the HMS cost and reliability, and was tested using three case studies involving different numbers of houses.

In [113], Ou and Hong proposed an MG hybrid PV-wind- fuel cell (FC) system-based stand-alone power supply. The performance of the PV generation source was analysed using an improved particle swarm optimization method (PSO) and a general regression neural network (GRNN). The optimal speed that the turbine should have attained to harness the most power from the wind was determined using a high-performance on-line training radial basis function network-sliding mode algorithm (RBFNSM). An intelligent controller was created that

Table 6
Case studies of optimisation applied to MGs integrated with renewable energy systems.

Hybrid design	Mode of operation	Factors considered for optimisation	Optimisation technique	Merits and demerits of optimisation technique	Title of the article	Ref.
PV, WT, FC, and diesel generators	On-grid	Cost, design	Monte Carlo simulation	*Merits:*- Simulations dealing with stochastic events are a part of the Monte Carlo method -Appropriately addresses uncertainty variables - Any solution chosen is completely independent of any previous decision and its outcomes *Demerits*: - The process requires complex calculations [117]	Optimal bidding strategy for microgrids considering renewable energy and building thermal dynamics	[111]
PV, WT, biomass, and battery	Off-grid	Cost, reliability, size	ABC	*Merits:* - Simplicity and the flexibility to explore local solutions - Ease of use in the implementation *Demerits:* - Ineffective for resolving complex problems [118]	Feasibility study of an islanded microgrid in rural area consisting of PV, wind, biomass and battery energy storage system.	[85]
PV, WT, and battery	On- and off-grid	Size	MILP	*Merits:* - A potent and flexible method for solving difficult problems - Accurate and effective in finding the optimal solution to the linear problem *Demerits:* - Nonlinear effects cannot be considered, and all time periods must be considered simultaneously [119]	Sizing and analysis of renewable energy and battery systems in residential microgrids	[112]
PV, WT, and battery	Off-grid	Power quality	Fuzzy logic and PSO	*Merits:* -The most important advantages of PSO is the simplicity of both concept and application level, and its fast convergence rate and ease of use. - Supports multi-objective optimisation problems *Demerits:* -Difficulties in controlling diversity when used as a multi-objective optimisation[120]	Intelligent frequency control in an AC microgrid: Online PSO-based fuzzy tuning approach	[96]
PV, WT, diesel, and battery	Off-grid	Cost, size, reliability	MOSaDE	*Merits:* -The good convergence ability -An effective approach to handle multi-objective optimisation problems *Demerits:* -computational complexity [121]	Optimal sizing of PV/wind/diesel hybrid microgrid system using multi-objective self-adaptive differential evolution algorithm	[100]
PV, WT,(FC), SVAR, and intelligent power controller	Off grid	Max power, control	PSO,GRNN, RBFNSM	-GRNN's merits include its accuracy and short training process. -The drawbacks of GRNN include its potentially enormous size, which would make it computationally expensive [122] - The merits of RBFNSM are its reliability, speed, and effectiveness. - Effective in solving classification and regression problems -RBFNSM has disadvantages in that the computational time is expensive when dealing with a high number of data points [123]	Dynamic operation and control of microgrid hybrid power systems	[113]
PV,WT,diesel, and battery	Off-grid	Emissions, LCC and MRP	GA	*Merits*: -Easy to understand the concept - Supports multi-objective optimisation problems *Demerits*: - Time consuming comparing to the PSO - Complicated computations [117]	Optimal sizing, operating strategy and operational experience of a stand-alone microgrid on Dongfushan Island	[114]
PV- thermal / WT, micro-turbine/diesel, electrical energy storage, and thermal energy storage	Off-grid	LPSP, COE, RF	E-PSO	*Merits:* -EPSO joins together the characteristics of evolutionary algorithms and particle swarm algorithm -EPSO can deal effectively with a diversity of objective functions	Optimal sizing and techno-economic analysis of energy- and cost-efficient standalone multi-carrier microgrid	[115]

(*continued on next page*)

Table 6 (*continued*)

Hybrid design	Mode of operation	Factors considered for optimisation	Optimisation technique	Merits and demerits of optimisation technique	Title of the article	Ref.
				-Very successful in solving the power system optimisation problem -EPSO has better convergence characteristics than the conventional PSO [124]		
PV	On-grid	Power prediction	Hybrid WT-PSO-SVM	Merits: -SVM algorithms involves solving convex and non-convex problems Demerits:-SVM algorithm is not suitable for large data sets -SVM model is computationally expensive and time consuming -To enhance the ability of classification and optimise parameters, SVM has been integrated with other advanced optimisation algorithms [125]	Short-term photovoltaic solar power forecasting using a hybrid wavelet-PSO-SVM model based on SCADA and meteorological information	[107]
PV, WT, PAFC, and battery	On-grid	EOM	GSA	*Merits:* -Simple implementation -GSA is good at finding the global optimum - Shows good results for non-linear optimisation problems *Demerits*: -Slow convergence speed and getting stuck in local minima in the last iterations -The GSA operators are complex [126]	Probabilistic energy and operation management of a microgrid containing wind/photovoltaic/fuel cell generation and energy storage devices based on point estimate method and self-adaptive gravitational search algorithm	[97]
Biomass, natural gas generator, and diesel generators	Standalone	Economic evaluation	HOMER	*Merits:* -Homer is a powerful tool for developing and sizing HRES components by utilizing techno-economic analysis. -Very thorough results for analysis and assessment. -Provide a realistic, regularly updated library of components *Demerits:* - If critical values or sizes are missed, HOMER won't attempt to guess them. -Time and detailed input data are required [127]	Integration of renewable energy in microgrids coordinated with demand response resources: Economic evaluation of a biomass gasification plant by Homer Simulator	[77]
PV, WT, and pumped storage	Standalone	LPSP, COE, LCC	GA	*Merits*: -Easy to understand the concept - Supports multi-objective optimisation problems *Demerits*: - Time consuming comparing to the PSO - Complicated computations [117]	Optimal design of an autonomous solar-wind-pumped storage power supply system	[116]

Key: PV = photovoltaic, WT = wind turbine, ABCA = artificial bee colony algorithm, MILP = mixed integer linear programming, PSO = particle swarm optimisation, MOSaDE = multi-objective self-adaptive differential evolution, GRNN = general regression neural network, RBFNSM = radial basis function network-sliding mode, GA = genetic algorithm, E-PSO = evolutionary particle swarm optimisation, FC = fuel cell, SVAR = static var compensator, LPSP = loss of power supply probability, COE = cost of electricity, RF = renewable factor, LCC = life cycle cost, MRP = maximisation of renewable energy source penetration, WT-PSO-SVM = wavelet transformation, particle swarm optimisation, and support vector machine, PAFC = phosphoric acid fuel cell, EOM = energy and operation management, GSA = gravitational search algorithm, MOGA = multi-objective genetic algorithm, HOMER = hybrid optimisation of multiple energy resources.

was composed of an RBFNSM and a GRNN for MPPT (maximum power point tracking) control in order to produce a quick and stable response for real power control.

A genetic algorithm (GA)-based method was used to solve the sizing optimisation problem with multiple objectives, including minimisation of the LCC, maximisation of RES penetration and minimisation of pollutant emissions. The proposed method was applied to the design and development of a real MG system on Dongfushan Island, Zhejiang Province, China, consisting of wind turbine generators, solar panels, diesel generators and battery storage units [114]. The optimal size and other technical and financial features of a standalone multi-carrier an MG integrated with a PV-thermal/small turbine WT/diesel system were analysed by Lorestani et al. in [115]. The optimisation problem was solved by an evolutionary particle swarm optimisation (E-PSO). The outcomes showed that the suggested arrangement was effective in terms of its financial, energy-efficient, dependable, and environmental qualities.

PV power generation is affected by changes in weather and solar radiation, which pose a significant barrier to integrating PV power into the grid. A hybrid prediction model that combined wavelet transformation, particle swarm optimisation, and a support vector machine (Hybrid WT-PSO-SVM) was proposed by Eseye et al. for short-term (one day ahead) power generation prediction for a real MG-PV system [107]. Niknam et al. [97] reported on the application of a self-adaptive optimisation algorithm based on the gravitational search algorithm (GSA) to identify the optimal energy management of a hybrid PV-WT-PAFC-battery MG in an uncertain environment. Their approach considered uncertainties in load demand, market prices, and the availability of electrical power from wind farms and solar systems [77]. Montuori et al. used HOMER software to carry out an economic evaluation of an MG supplied by a biomass gasification power plant and isolated from the grid, and compared this to alternative generation technologies (diesel generator/gas generator). A techno-economic analysis of a standalone hybrid solar-wind-pumped storage system for an isolated MG was presented by Ma et al. [116]. To find the optimal system configuration, a GA optimisation was used. The results showed that the COE for the system was marginally lower than that of a solar-only system and much lower than that of a wind-only system, owing to the high cost of wind turbines. Pumped storage systems based on renewable energy were found to be able to supply stable and uninterrupted power to remote areas.

Table 6 shows that extensive research has gone into determining the optimal sizes of HMGS components to ensure that all load requirements are met with a low COE and the highest level of reliability. The optimisation of HMGSs is crucial because it offers a variety of design options and operational scenarios that can help academics and professionals choose the optimal system configuration. Given that the cost of PV has been steadily declining over the past 30 years as a result of technological advancements in the solar panel sector, solar PV is used as one of the sources of the hybrid system in the majority of HMGSs. Due to environmental concerns, fluctuating fossil fuel prices, and the fact that the cost of PV and wind has decreased by 85 % and 56 %, respectively, over the past 10 years, PV-wind is the most well-known hybrid RE system. This discovery has motivated researchers to carry out numerous investigations in this field.

4. Conclusions

This paper has presented a review of research developments related to three themes: microgrids (MGs), renewable energy (RE), and the optimisation of these systems. The systematic literature review was based on a SciMAT bibliometric analysis of the evolution of the selected field of research between 2005 and 2021, using publications available from Scopus. The trends were analysed based on an overview and a more specific analysis of three different phases of the period under review (2005–2011, 2012–2015, and 2016–2021).

The analysis revealed that multi-objective optimisation and MGs have been prominent areas of research, particularly during the last five years. Since 2016, there has also been a notable rise in the number of articles on these subjects that have been published in international journals. Overlapping graphs by period showed significant developments in the number of new and transitional keywords, with an increase of 406 % between the first and last periods, indicating that research on each of the topics in the analysed domain is in a state of continuous development. Strategic diagrams were created to analyse the changes in the field of research over the three periods studied, and these indicated an emphasis on developments in the technical and economic evaluation of hybrid microgrid systems (HMGSs). The first period (2005–2011) contained a large number of documents focusing on two topics (smart grids and optimisation). Within this period, the US (DOE) Smart Grid Research and Development Program stated that MGs were basic building blocks for the smart grid. Following the first workshop held by the DOE in August 2011 that brought together experts and practitioners to identify priorities in the field of research and development of MGs, the end of this period is therefore considered to be the beginning of the development that occurs in research that studies the evolution of the MG. The second period (2012–2015) contained the largest number of documents focusing on MGs and the distribution energy resources. The most recent period (2016–2021) indicated interest and a very large increase in the number of studies on MGs and multi-objective optimisation. The use of smart grids rather than traditional grids has attracted research attention due to the increasing interest in the efficiency, reliability, economy, and sustainability of the production and distribution of electricity. This was evident in the strategic plan for each period under study as the subject of smart grids was covered in a significant number of documents with numerous citations.

Due to the lower cost and new developments in RESs components, the rise in the number of research studies evaluating these networks, and the significant advancements in algorithms for solving optimisation problems related to these systems, this study has demonstrated that MGs integrated with RESs represent the future of power systems.

5. Suggestions for further studies

1. The authors propose that the assessment of HMGs should be expanded to include economic, technical, environmental, and social aspects as well as an examination of the use of these systems in developing countries experiencing energy shortages. To draw the attention of the governments of these nations to the need for investment in renewable energy.
2. This analysis found that the literature on multi-objective optimisation used to optimise HMGSs is very popular. We advise conducting a comparison study on single- and multi-objective optimisation as well as software tools which used to evaluate HMGSs.
3. The literature on the subjects of MGs and multi-objective optimisation is of particular interest in the third period. The authors advise conducting a bibliometric analysis study to examine the evolution of these themes.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] Sen S, Ganguly S, Das A, Sen J, Dey S. Renewable energy scenario in India: Opportunities and challenges; 2015. doi: 10.1016/j.jafrearsci.2015.06.002.
[2] "Renewable Power – Analysis - IEA." https://www.iea.org/reports/renewable-power (accessed Mar. 11, 2022).
[3] "Solar PV – Analysis - IEA." https://www.iea.org/reports/solar-pv (accessed Jan. 04, 2022).
[4] IREA. Renewable power generation costs in 2020; 2020.
[5] Paliwal P, Patidar NP, Nema RK. Planning of grid integrated distributed generators: A review of technology, objectives and techniques. Renew Sustain Energy Rev 2014;40:557–70. https://doi.org/10.1016/J.RSER.2014.07.200.
[6] Lazarov VD, Zarkov Z, Bochev I. Hybrid power systems with renewable energy sources-types, structures, trends for research and development, no. May 2014, 2005, [Online]. Available: https://www.researchgate.net/publication/236012467.
[7] Upadhyay S, Sharma MP. A review on configurations, control and sizing methodologies of hybrid energy systems. Renew Sustain Energy Rev 2014;38:47–63. https://doi.org/10.1016/J.RSER.2014.05.057.
[8] Yang H, Lu L, Zhou W. A novel optimization sizing model for hybrid solar-wind power generation system. Sol Energy 2007;81(1):76–84. https://doi.org/10.1016/J.SOLENER.2006.06.010.
[9] Deshmukh MK, Deshmukh SS. Modeling of hybrid renewable energy systems. Renew Sustain Energy Rev 2008;12(1):235–49. https://doi.org/10.1016/J.RSER.2006.07.011.
[10] de Oliveira e Silva G, Hendrick P. Photovoltaic self-sufficiency of Belgian households using lithium-ion batteries, and its impact on the grid. Appl Energy, 195, pp. 786–799, Jun. 2017, doi: 10.1016/J.APENERGY.2017.03.112.
[11] Schram WL, Lampropoulos I, van Sark WGJHM. Photovoltaic systems coupled with batteries that are optimally sized for household self-consumption: Assessment of peak shaving potential. Appl Energy Aug. 2018;223:69–81. https://doi.org/10.1016/J.APENERGY.2018.04.023.
[12] Sharma S, Bhattacharjee S, Bhattacharya A. Grey wolf optimisation for optimal sizing of battery energy storage device to minimise operation cost of microgrid. IET Gener Transm Distrib 2016;10(3):625–37. https://doi.org/10.1049/IET-GTD.2015.0429.
[13] Fathima AH, Palanisamy K. Optimization in microgrids with hybrid energy systems – A review. Renew Sustain Energy Rev 2015;45:431–46. https://doi.org/10.1016/J.RSER.2015.01.059.

[14] Kallio S, Siroux M. Hybrid renewable energy systems based on micro-cogeneration. Energy Rep 2022;8:762–9. https://doi.org/10.1016/J.EGYR.2021.11.158.
[15] Chang KH, Lin G. Optimal design of hybrid renewable energy systems using simulation optimization. Simul Model Pract Theory 2015;52:40–51. https://doi.org/10.1016/J.SIMPAT.2014.12.002.
[16] Guangqian D, Bekhrad K, Azarikhah P, Maleki A. A hybrid algorithm based optimization on modeling of grid independent biodiesel-based hybrid solar/wind systems. Renew Energy 2018;122:551–60. https://doi.org/10.1016/J.RENENE.2018.02.021.
[17] Maheri A. A critical evaluation of deterministic methods in size optimisation of reliable and cost effective standalone hybrid renewable energy systems. Reliab Eng Syst Saf 2014;130:159–74. https://doi.org/10.1016/J.RESS.2014.05.008.
[18] Ogunjuyigbe ASO, Ayodele TR, Akinola OA. Optimal allocation and sizing of PV/Wind/Split-diesel/Battery hybrid energy system for minimizing life cycle cost, carbon emission and dump energy of remote residential building. Appl Energy 2016;171:153–71. https://doi.org/10.1016/J.APENERGY.2016.03.051.
[19] Roberts JJ, Marotta Cassula A, Silveira JL, da Costa Bortoni E, Mendiburu AZ. Robust multi-objective optimization of a renewable based hybrid power system. Appl Energy 2018;223:52–68.
[20] Zia MF, Elbouchikhi E, Benbouzid M. Microgrids energy management systems: A critical review on methods, solutions, and prospects. Appl Energy 2018;222:1033–55. https://doi.org/10.1016/J.APENERGY.2018.04.103.
[21] Ton DT, Smith MA. The U.S. department of energy's microgrid initiative. Electr J 2012;25(8):84–94. https://doi.org/10.1016/j.tej.2012.09.013.
[22] Azaza M, Wallin F. Multi objective particle swarm optimization of hybrid micro-grid system: A case study in Sweden. Energy 2017;123:108–18. https://doi.org/10.1016/j.energy.2017.01.149.
[23] Borhanazad H, Mekhilef S, Gounder Ganapathy V, Modiri-Delshad M, Mirtaheri A. Optimization of micro-grid system using MOPSO. Renew Energy 2014;71:295–306. https://doi.org/10.1016/j.renene.2014.05.006.
[24] Soshinskaya M, Crijns-Graus WHJ, Guerrero JM, Vasquez JC. Microgrids: Experiences, barriers and success factors. Renew Sustain Energy Rev 2014;40:659–72. https://doi.org/10.1016/J.RSER.2014.07.198.
[25] Bajpai P, Dash V. Hybrid renewable energy systems for power generation in stand-alone applications: A review. Renew Sustain Energy Rev 2012;16(5):2926–39. https://doi.org/10.1016/J.RSER.2012.02.009.
[26] Kharrich M, Mohammed OH, Alshammari N, Akherraz M. Multi-objective optimization and the effect of the economic factors on the design of the microgrid hybrid system. Sustain Cities Soc 2021;65:102646.
[27] Dougier N, Garambois P, Gomand J, Roucoules L. Multi-objective non-weighted optimization to explore new efficient design of electrical microgrids. Appl Energy 2021;304:117758. https://doi.org/10.1016/J.APENERGY.2021.117758.
[28] Zarate-Perez E, Rosales-Asensio E, González-Martínez A, de Simón-Martín M, Colmenar-Santos A. Battery energy storage performance in microgrids: A scientific mapping perspective. Energy Rep 2022;8:259–68. https://doi.org/10.1016/J.EGYR.2022.06.116.
[29] Roslan MF, Hannan MA, Ker PJ, Mannan M, Muttaqi KM, Mahlia TI. Microgrid control methods toward achieving sustainable energy management: A bibliometric analysis for future directions. J Clean Prod 2022;348:131340. https://doi.org/10.1016/J.JCLEPRO.2022.131340.
[30] Reza MS, Rahman N, Wali SB, Hannan MA, Ker PJ, Rahman SA, et al. Optimal algorithms for energy storage systems in microgrid applications: an analytical evaluation towards future directions. IEEE Access 2022;10:10105–23.
[31] Kavadias KA, Triantafyllou P. Hybrid renewable energy systems' optimisation. A review and extended comparison of the most-used software tools; 2021. doi: 10.3390/en14248268.
[32] Shaikh PH, Shaikh A, Memon ZA, Lashari AA, Leghari ZH. Microgrids: A review on optimal hybrid technologies, configurations, and applications. Int J Energy Res 2021;45(9):12564–97.
[33] Dawoud SM, Lin X, Okba MI. Hybrid renewable microgrid optimization techniques: A review. Renew Sustain Energy Rev Feb. 2018;82:2039–52. https://doi.org/10.1016/J.RSER.2017.08.007.
[34] van Nunen K, Li J, Reniers G, Ponnet K. Bibliometric analysis of safety culture research. Saf Sci Oct. 2018;108:248–58. https://doi.org/10.1016/J.SSCI.2017.08.011.
[35] "Combining mapping and citation analysis for evaluative bibliometric purposes: A bibliometric study - Noyons - 1999 - Journal of the American Society for Information Science - Wiley Online Library." https://asistdl.onlinelibrary.wiley.com/doi/abs/10.1002/(SICI)1097-4571(1999)50:2%3C115::AID-ASI3%3E3.0.CO;2-J (accessed Mar. 21, 2022).
[36] Osareh F. Bibliometncs, citation analysis and co-cüation analysis, vol. 46, pp. 149–158; 1996.
[37] Roig-Tierno N, Gonzalez-Cruz TF, Llopis-Martinez J. An overview of qualitative comparative analysis: A bibliometric analysis. J Innov Knowl 2017;2(1):15–23. https://doi.org/10.1016/J.JIK.2016.12.002.
[38] Small H. Visualizing science by citation mapping; 1999, doi: 10.1002/(SICI)1097-4571(1999)50:93.3.CO;2-7.
[39] Albort-Morant G, Ribeiro-Soriano D. A bibliometric analysis of international impact of business incubators. J Bus Res 2016;69(5):1775–9. https://doi.org/10.1016/J.JBUSRES.2015.10.054.
[40] Bhimani H, Mention AL, Barlatier PJ. Social media and innovation: A systematic literature review and future research directions. Technol Forecast Soc Change 2019;144:251–69. https://doi.org/10.1016/J.TECHFORE.2018.10.007.
[41] Antman EM, Lau J, Kupelnick B, Mosteller F, Chalmers TC, Chalmers H. A comparison of results of meta-analyses of randomized control trials and recommendations of clinical experts treatments for myocardial infarction. Accessed: Mar. 27, 2022. [Online]. Available: http://jama.jamanetwork.com/.
[42] Martínez-Aires MD, López-Alonso M, Martínez-Rojas M. Building information modeling and safety management: A systematic review. Saf Sci Jan. 2018;101:11–8. https://doi.org/10.1016/J.SSCI.2017.08.015.
[43] Savaget P, Geissdoerfer M, Kharrazi A, Evans S. The theoretical foundations of sociotechnical systems change for sustainability: A systematic literature review. J Clean Prod Jan. 2019;206:878–92. https://doi.org/10.1016/J.JCLEPRO.2018.09.208.
[44] Theisen C, Munaiah N, Al-Zyoud M, Carver JC, Meneely A, Williams L. Attack surface definitions: A systematic literature review. Inf Softw Technol Dec. 2018;104:94–103. https://doi.org/10.1016/J.INFSOF.2018.07.008.
[45] Gupta S, Rajiah P, Middlebrooks EH, Baruah D, Carter BW, Burton KR, et al. Systematic review of the literature: best practices. Acad Radiol 2018;25(11):1481–90.
[46] Binali E. Managing airports in non-aviation related disasters: A systematic literature review Abdussamet Polater; 2018, doi: 10.1016/j.ijdrr.2018.05.026.
[47] Ruhlandt RWS. The governance of smart cities: A systematic literature review. Cities 2018;81:1–23. https://doi.org/10.1016/J.CITIES.2018.02.014.
[48] Kitchenham B. Guidelines for performing systematic literature reviews in software engineering; 2007. Accessed: Mar. 21, 2022. [Online]. Available: https://www.researchgate.net/publication/302924724.
[49] Liberati A, Altman DG, Tetzlaff J, Mulrow C, Gotzsche PC, Ioannidis JPA, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. BMJ 2009;339(jul21 1):b2700–.
[50] Callon M, Courtial J-P, Turner WA, Bauin S. From translations to problematic networks: An introduction to co-word analysis. Social Sci Inform 1983;22(2):191–235.
[51] Hirsch JE. An index to quantify an individual's scientific research output. Proc Natl Acad Sci U S A 2005;102(46):16569–72. https://doi.org/10.1073/PNAS.0507655102/ASSET/8BAD4608-B7BF-4215-8D0A-5C29F690AE48/ASSETS/GRAPHIC/ZPQ0420598740002.JPEG.
[52] Martínez MA, Cobo MJ, Herrera M, Herrera-Viedma E. Analyzing the scientific evolution of social work using science mapping. Res Social Work Practice 2015;25(2):257–77. https://doi.org/10.1177/1049731514522101.
[53] Oakleaf M. Writing information literacy assessment plans: a guide to best practice. Commun Inf Lit 2010;3(2):4. https://doi.org/10.15760/comminfolit.2010.3.2.73.
[54] Cobo MJ, López-Herrera AG, Herrera-Viedma E, Herrera F. An approach for detecting, quantifying, and visualizing the evolution of a research field: A practical application to the Fuzzy Sets Theory field. J Informetr 2011;5(1):146–66. https://doi.org/10.1016/J.JOI.2010.10.002.
[55] Fernández-González JM, Díaz-López C, Martín-Pascual J, Zamorano M. Recycling organic fraction of municipal solid waste: systematic literature review and bibliometric analysis of research trends. Sustain 2020;12(11):4798. https://doi.org/10.3390/SU12114798.
[56] Xie H, Zhang Y, Duan K. Evolutionary overview of urban expansion based on bibliometric analysis in Web of Science from 1990 to 2019. Habitat Int Jan. 2020;95:102100. https://doi.org/10.1016/J.HABITATINT.2019.102100.
[57] Carpio M, González Á, González M, Verichev K. Influence of pavements on the urban heat island phenomenon: A scientific evolution analysis. Energy Build 2020;226:110379. https://doi.org/10.1016/J.ENBUILD.2020.110379.
[58] López-Alonso M, Martín-Morales M, Martínez-Echevarría MJ, Agrela F, Zamorano M. Residual biomasses as aggregates applied in cement-based materials. Waste Byprod Cem Mater 2021;1:89–137. https://doi.org/10.1016/B978-0-12-820549-5.00011-5.
[59] Casado-Aranda L-A, Sánchez-Fernández J, Viedma-del-Jesús MI. Analysis of the scientific production of the effect of COVID-19 on the environment: A bibliometric study. Environ Res 2021;193:110416.
[60] Díaz-López C, Carpio M, Martín-Morales M, Zamorano M. Analysis of the scientific evolution of sustainable building assessment methods. Sustain Cities Soc Aug. 2019;49:101610. https://doi.org/10.1016/J.SCS.2019.101610.
[61] David TM, Silva Rocha Rizol PM, Guerreiro Machado MA, Buccieri GP. Future research tendencies for solar energy management using a bibliometric analysis, 2000–2019. Heliyon 2020;6(7):e04452. https://doi.org/10.1016/J.HELIYON.2020.E04452.
[62] Aparicio G, Iturralde T, Maseda A. Conceptual structure and perspectives on entrepreneurship education research: A bibliometric review. Eur Res Manag Bus Econ Sep. 2019;25(3):105–13. https://doi.org/10.1016/J.IEDEEN.2019.04.003.
[63] Santana M, Cobo MJ. What is the future of work? A science mapping analysis. Eur Manag J 2020;38(6):846–62. https://doi.org/10.1016/J.EMJ.2020.04.010.
[64] López-Robles JR, Rodríguez-Salvador M, Gamboa-Rosales NK, Ramirez-Rosales S, Cobo MJ. The last five years of big data research in economics, econometrics and finance: identification and conceptual analysis. Procedia Comput Sci 2019;162:729–36. https://doi.org/10.1016/J.PROCS.2019.12.044.
[65] Kipper LM, Iepsen S, Dal Forno AJ, Frozza R, Furstenau L, Agnes J, et al. Scientific mapping to identify competencies required by industry 4.0. Technol Soc 2021;64:101454.
[66] Sharifi A. Urban sustainability assessment: An overview and bibliometric analysis. Ecol Indic 2021;121:107102. https://doi.org/10.1016/J.ECOLIND.2020.107102.
[67] Callon M, Courtial JP, Laville F. Co-word analysis as a tool for describing the network of interactions between basic and technological research: The case of polymer chemsitry. Scientometrics 1991;22(1):155–205.

[68] Coulter SKN, Monarch I. (PDF) software engineering as seen through its research literature: a study in co-word analysis. https://www.researchgate.net/publication/220433686_Software_Engineering_as_Seen_through_Its_Research_Literature_A_Study_in_Co-Word_Analysis (accessed Mar. 21, 2022).
[69] Agency IE, et al. SDG indicator metadata; 2021.
[70] Tsikalakis AG, Hatziargyriou ND. Centralized control for optimizing microgrids operation. IEEE Power Energy Soc Gen Meet 2011. https://doi.org/10.1109/PES.2011.6039737.
[71] Sortomme E, El-Sharkawi MA. Optimal power flow for a system of microgrids with controllable loads and battery storage; optimal power flow for a system of microgrids with controllable loads and battery storage; 2009.
[72] Yu X, Khambadkone AM, Wang H, Terence STS. Control of parallel-connected power converters for low-voltage microgrid—Part I: A hybrid control architecture. IEEE Trans Power Electron 2010;25(12):2962–70.
[73] Boroyevich D, Cvetković I, Dong D, Burgos R, Wang F, Lee F. Future electronic power distribution systems - A contemplative view. Proc Int Conf Optim Electr Electron Equipment, OPTIM 2010:1369–80. https://doi.org/10.1109/OPTIM.2010.5510477.
[74] Chakraborty S, Weiss MD, Simões MG. Distributed intelligent energy management system for a single-phase high-frequency AC microgrid. IEEE Trans Ind Electron 2007;54(1):97–109. https://doi.org/10.1109/TIE.2006.888766.
[75] Chung I-Y, Liu W, Cartes DA, Collins EG, Moon S-I. Control methods of inverter-interfaced distributed generators in a microgrid system. IEEE Trans Ind Appl 2010;46(3):1. https://doi.org/10.1109/TIA.2010.2044970.
[76] Kanchev H, Colas F, Lazarov V, Francois B. Emission reduction and economical optimization of an urban microgrid operation including dispatched PV-based active generators. IEEE Trans Sustain Energy 2014;5(4):1397–405.
[77] Montuori L, Alcázar-Ortega M, Álvarez-Bel C, Domijan A. Integration of renewable energy in microgrids coordinated with demand response resources: Economic evaluation of a biomass gasification plant by Homer Simulator. Appl Energy 2014;132:15–22. https://doi.org/10.1016/J.APENERGY.2014.06.075.
[78] Zhang Yu, Gatsis N, Giannakis GB. Robust energy management for microgrids with high-penetration renewables. IEEE Trans Sustain Energy 2013;4(4):944–53.
[79] Chaouachi A, Kamel RM, Andoulsi R, Nagasaka K. Multiobjective intelligent energy management for a microgrid. IEEE Trans Ind Electron 2013;60(4):1688–99.
[80] Zhang D, Shah N, Papageorgiou LG. Efficient energy consumption and operation management in a smart building with microgrid. Energy Convers Manag Oct. 2013;74:209–22. https://doi.org/10.1016/J.ENCONMAN.2013.04.038.
[81] Conti S, Nicolosi R, Rizzo SA, Zeineldin HH. Optimal dispatching of distributed generators and storage systems for MV islanded microgrids. IEEE Trans Power Deliv 2012;27(3):1243. https://doi.org/10.1109/TPWRD.2012.2194514.
[82] Liao GC. Solve environmental economic dispatch of Smart MicroGrid containing distributed generation system – Using chaotic quantum genetic algorithm. Int J Electr Power Energy Syst 2012;43(1):779–87. https://doi.org/10.1016/J.IJEPES.2012.06.040.
[83] Chen YH, Lu SY, Chang YR, Lee TT, Hu MC. Economic analysis and optimal energy management models for microgrid systems: A case study in Taiwan. Appl Energy 2013;103:145–54. https://doi.org/10.1016/J.APENERGY.2012.09.023.
[84] Nguyen TA, Crow ML. Stochastic optimization of renewable-based microgrid operation incorporating battery operating cost. IEEE Trans Power Syst 2016;31(3):2289–96.
[85] Singh S, Singh M, Kaushik SC. Feasibility study of an islanded microgrid in rural area consisting of PV, wind, biomass and battery energy storage system. Energy Convers Manag 2016;128:178–90. https://doi.org/10.1016/J.ENCONMAN.2016.09.046.
[86] Korkas CD, Baldi S, Michailidis I, Kosmatopoulos EB. Occupancy-based demand response and thermal comfort optimization in microgrids with renewable energy sources and energy storage. Appl Energy 2016;163:93–104. https://doi.org/10.1016/J.APENERGY.2015.10.140.
[87] Wang Z, Yu X, Mu Y, Jia H, Jiang Q, Wang X. Peer-to-Peer energy trading strategy for energy balance service provider (EBSP) considering market elasticity in community microgrid. Appl Energy 2021;303:117596. https://doi.org/10.1016/J.APENERGY.2021.117596.
[88] Fu Y, Zhang Z, Li Z, Mi Y. Energy management for hybrid AC/DC distribution system with microgrid clusters using non-cooperative game theory and robust optimization. IEEE Trans Smart Grid 2020;11(2):1510–25. https://doi.org/10.1109/TSG.2019.2939586.
[89] Abdolrasol MGM, Hannan MA, Mohamed A, Amiruldin UAU, Abidin IBZ, Uddin MN. An optimal scheduling controller for virtual power plant and microgrid integration using the binary backtracking search algorithm. IEEE Trans on Ind Applicat 2018;54(3):2834–44.
[90] Heydari A, Astiaso Garcia D, Keynia F, Bisegna F, De Santoli L. A novel composite neural network based method for wind and solar power forecasting in microgrids. Appl Energy 2019;251:113353.
[91] Masoumi A, Ghassem-zadeh S, Hosseini SH, Ghavidel BZ. Application of neural network and weighted improved PSO for uncertainty modeling and optimal allocating of renewable energies along with battery energy storage. Appl Soft Comput 2020;88:105979. https://doi.org/10.1016/J.ASOC.2019.105979.
[92] Wang T, He X, Deng T. Neural networks for power management optimal strategy in hybrid microgrid. Neural Comput & Applic 2019;31(7):2635–47.
[93] Ullah K, Hafeez G, Khan I, Jan S, Javaid N. A multi-objective energy optimization in smart grid with high penetration of renewable energy sources. Appl Energy 2021;299:117104. https://doi.org/10.1016/J.APENERGY.2021.117104.
[94] "Smart Grids – Analysis - IEA." https://www.iea.org/reports/smart-grids (accessed Mar. 28, 2022).
[95] Liang H, Tamang AK, Zhuang W, Shen XS. Stochastic information management in smart grid. IEEE Commun Surv Tutorials 2014;16(3):1746–70.
[96] Bevrani H, Habibi F, Babahajyani P, Watanabe M, Mitani Y. Intelligent frequency control in an AC microgrid: Online PSO-based fuzzy tuning approach. IEEE Trans Smart Grid 2012;3(4):1935–44. https://doi.org/10.1109/TSG.2012.2196806.
[97] Niknam T, Golestaneh F, Malekpour A. Probabilistic energy and operation management of a microgrid containing wind/photovoltaic/fuel cell generation and energy storage devices based on point estimate method and self-adaptive gravitational search algorithm. Energy 2012;43(1):427–37. https://doi.org/10.1016/J.ENERGY.2012.03.064.
[98] Zhang D, Shah N, Papageorgiou LG. Efficient energy consumption and operation management in a smart building with microgrid. Energy Convers Manag 2013;74:209–22. https://doi.org/10.1016/J.ENCONMAN.2013.04.038.
[99] Liang H, Zhuang W. Stochastic modeling and optimization in a microgrid: a survey. Energies 2014; 7: 2027–2050, doi: 10.3390/EN7042027.
[100] Ramli MAM, Bouchekara HREH, Alghamdi AS. Optimal sizing of PV/wind/diesel hybrid microgrid system using multi-objective self-adaptive differential evolution algorithm. Renew Energy 2018;121:400–11. https://doi.org/10.1016/J.RENENE.2018.01.058.
[101] Shen J, Jiang C, Liu Y, Wang X. A microgrid energy management system and risk management under an electricity market environment. IEEE Access 2016;4:2349–56. https://doi.org/10.1109/ACCESS.2016.2555926.
[102] Schreiber M. Restricting the h-index to a publication and citation time window: A case study of a timed Hirsch index. J Informetr 2015;9(1):150–5. https://doi.org/10.1016/J.JOI.2014.12.005.
[103] Wang Z, Chen B, Wang J, Kim J. Decentralized energy management system for networked microgrids in grid-connected and islanded modes. IEEE Trans Smart Grid 2016;7(2):1097–105. https://doi.org/10.1109/TSG.2015.2427371.
[104] Zhao Bo, Wang X, Lin Da, Calvin MM, Morgan JC, Qin R, et al. Energy management of multiple microgrids based on a system of systems architecture. IEEE Trans Power Syst 2018;33(6):6410–21.
[105] Liu Y, Li Y, Gooi HB, Jian Ye, Xin H, Jiang X, et al. Distributed robust energy management of a multimicrogrid system in the real-time energy market. IEEE Trans Sustain Energy 2019;10(1):396–406.
[106] Ju C, Wang P, Goel L, Xu Y. A two-layer energy management system for microgrids with hybrid energy storage considering degradation costs. IEEE Trans Smart Grid 2018;9(6):6047–57. https://doi.org/10.1109/TSG.2017.2703126.
[107] Eseye AT, Zhang J, Zheng D. Short-term photovoltaic solar power forecasting using a hybrid Wavelet-PSO-SVM model based on SCADA and Meteorological information. Renew Energy 2018;118:357–67. https://doi.org/10.1016/J.RENENE.2017.11.011.
[108] Wang Y, Li F, Yu H, Wang Y, Qi C, Yang J, et al. Optimal operation of microgrid with multi-energy complementary based on moth flame optimization algorithm. Energy Sources Part A 2020;42(7):785–806.
[109] Luo Z, Wu Z, Li Z, Cai HY, Li BJ, Gu W. A two-stage optimization and control for CCHP microgrid energy management. Appl Therm Eng 2017;125:513–22. https://doi.org/10.1016/J.APPLTHERMALENG.2017.05.188.
[110] Zhang B, Li Q, Wang L, Feng W. Robust optimization for energy transactions in multi-microgrids under uncertainty. Appl Energy 2018;217:346–60. https://doi.org/10.1016/J.APENERGY.2018.02.121.
[111] Nguyen DT, Le LB. Optimal bidding strategy for microgrids considering renewable energy and building thermal dynamics. IEEE Trans Smart Grid 2014;5(4):1608–20. https://doi.org/10.1109/TSG.2014.2313612.
[112] Atia R, Yamada N. Sizing and analysis of renewable energy and battery systems in residential microgrids. IEEE Trans Smart Grid 2016;7(3):1204–13. https://doi.org/10.1109/TSG.2016.2519541.
[113] Ou TC, Hong CM. Dynamic operation and control of microgrid hybrid power systems. Energy 2014;66:314–23. https://doi.org/10.1016/J.ENERGY.2014.01.042.
[114] Zhao B, Zhang X, Li P, Wang K, Xue M, Wang C. Optimal sizing, operating strategy and operational experience of a stand-alone microgrid on Dongfushan Island. Appl Energy 2014;113:1656–66. https://doi.org/10.1016/J.APENERGY.2013.09.015.
[115] Lorestani A, Gharehpetian GB, Nazari MH. Optimal sizing and techno-economic analysis of energy- and cost-efficient standalone multi-carrier microgrid. Energy 2019;178:751–64. https://doi.org/10.1016/J.ENERGY.2019.04.152.
[116] Ma T, Yang H, Lu L, Peng J. Optimal design of an autonomous solar–wind-pumped storage power supply system. Appl Energy 2015;160:728–36. https://doi.org/10.1016/J.APENERGY.2014.11.026.
[117] Coello CAC, Lamont GB, Van Veldhuizen DA. Evolutionary algorithms for solving multi-objective problems; 2007.
[118] Yazdani D, Meybodi MR. A novel artificial bee colony algorithm for global optimization. Proc. 4th Int. Conf. Comput. Knowl. Eng. ICCKE; 2014. p. 443–448, Dec. 2014, doi: 10.1109/ICCKE.2014.6993393.
[119] Urbanucci L. Limits and potentials of mixed integer linear programming methods for optimization of polygeneration energy systems. Energy Procedia 2018;148:1199–205. https://doi.org/10.1016/J.EGYPRO.2018.08.021.
[120] Juneja M, Nagar SK. Particle swarm optimization algorithm and its parameters: A review. ICCCCM 2016–2nd IEEE Int. Conf. Control Comput. Commun Mater 2017. https://doi.org/10.1109/ICCCCM.2016.7918233.
[121] Wang Y-N, Wu L-H, Yuan X-F. Multi-objective self-adaptive differential evolution with elitist archive and crowding entropy-based diversity measure. Soft Comput 2010;14(3):193–209.
[122] Al-Mahasneh AJ, Anavatti SG, Garratt MA. Review of applications of generalized regression neural networks in identification and control of dynamic systems; 2018.

[123] Lin GF, Chen LH. A non-linear rainfall-runoff model using radial basis function network. J Hydrol 2004;289(1–4):1–8. https://doi.org/10.1016/J.JHYDROL.2003.10.015.

[124] Miranda V, Fonseca N. EPSO-evolutionary particle swarm optimization, a new algorithm with applications in power systems; 2002. doi: 10.1109/TDC.2002.1177567.

[125] Cervantes J, Garcia-Lamont F, Rodríguez-Mazahua L, Lopez A. A comprehensive survey on support vector machine classification: Applications, challenges and trends. Neurocomputing 2020;408:189–215. https://doi.org/10.1016/J.NEUCOM.2019.10.118.

[126] Rather SA, Bala PS. A holistic review on gravitational search algorithm and its hybridization with other optimization algorithms. Proc. 2019 3rd IEEE Int. Conf. Electr. Comput. Commun. Technol. ICECCT 2019, Feb. 2019, doi: 10.1109/ICECCT.2019.8869279.

[127] (PDF) Optimization of Renewable Energy Efficiency using HOMER. https://www.researchgate.net/publication/263765594_Optimization_of_Renewable_Energy_Efficiency_using_HOMER (accessed Sep. 02, 2022).